4.3.6

EE25BTECH11022 - sankeerthan

Question:passing through the points $(3, 4, -7)$ and $(1, -1, 6)$.
solution:let Passing through the points

$$\mathbf{A} = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix} \tag{1}$$

The direction vector of line passing through $\mathbf{A}$ and $\mathbf{B}$ is

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 1 - 3 \\ -1 - 4 \\ 6 - (-7) \end{pmatrix} \tag{2}$$

$$= \begin{pmatrix} -2 \\ -5 \\ 13 \end{pmatrix} \tag{3}$$

The equation of line is

$$r(t) = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix} + t \begin{pmatrix} -2 \\ -5 \\ 13 \end{pmatrix} \tag{4}$$

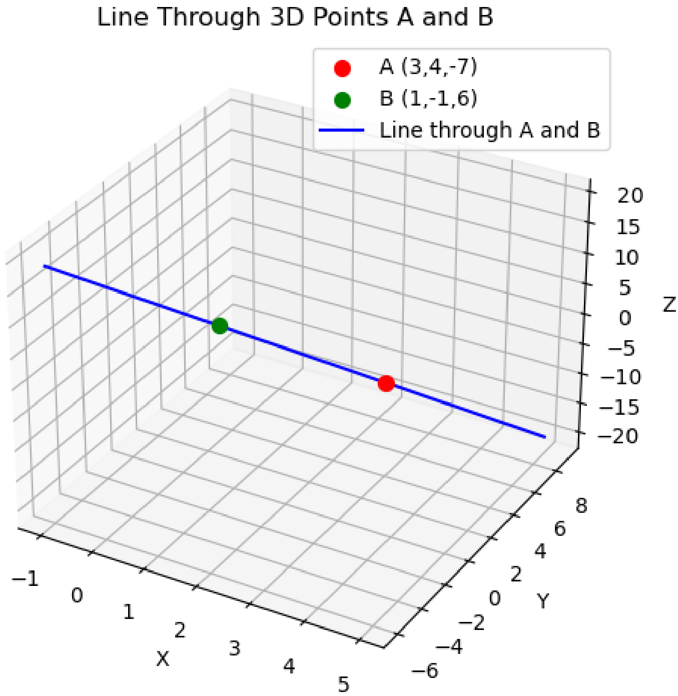
Line Through 3D Points A and B
A (3,4,-7)
B (1,-1,6)
Line through A and B
X
Y
Z
−1
0
1
2
3
4
5
−6
−4
−2
0
2
4
6
8
20
15
10
5
0
−5
−10
−15
−20

Fig. 0.